Vance Vanvolkenburgh
655.662 — Introduction to Healthcare Systems Engineering
Module 6 | M6A1: Problem Set — Kossiakoff Problem 13.2

Risk Reduction in the Advanced Development Phase

13.2 — What specific activities in the advanced development phase sometimes cause it to be referred to as a "risk reduction" phase? Give an example of each activity considering a real or hypothetical system.

Advanced development sits between a chosen concept and a committed design. Its purpose is not to build the system but to retire the uncertainty that would otherwise surface later, when correcting it is expensive. Six specific activities give the phase its other name, and the course materials state the objective plainly: the aim of prototyping and competition is **to reduce technical, cost and schedule risk**. Each activity below is paired with the class of risk it retires and an example from PD-RPM, the peritoneal dialysis remote patient monitoring system developed across this course.

1. Identifying and ranking the high-risk elements. The phase opens by deciding where the risk actually is, so that effort goes to the few elements that need it rather than being spread evenly. This retires **programmatic risk** — the risk of spending the phase on the wrong things. *Example:* a technology readiness survey of PD-RPM's twenty physical elements put most at TRL 9 — commercial cuffs, scales, workstations — and exactly one at TRL 6, the HG-1 home gateway with its cellular store-and-forward firmware. The companion risk, silent loss of home data delivery, scored 4 × 4 = 16. Advanced development was therefore scoped almost entirely around HG-1, and the commercial elements were left alone.

2. Requirements analysis and flow-down to a validated specification. Operational requirements are restated as testable system requirements, each tagged to the sub-function it bounds. This retires **verification risk**: ambiguity is found on paper rather than at acceptance test. *Example:* the PD-RPM operational requirement to transfer each completed treatment record within 30 minutes of session end flowed down to a pairing timeout of 10 seconds, a buffer write-confirm of 2 seconds, a 5-minute upload retry interval over twelve attempts, and an ingest-to-available latency of 90 seconds. Only once the budget was written out did it become visible that it left no allowance for a carrier re-registration, which routinely takes four minutes.

3. Trade studies among component alternatives. Engineering trade studies are the primary tool of this phase. Scoring alternatives against weighted, requirement-traced criteria retires **cost and schedule risk**, because the losing options are eliminated before anything is procured. *Example:* the PD-RPM connectivity trade study evaluated four alternatives against four criteria. The smartphone-tethered option, cheaper than the others by a factor of four, was disqualified on the data-completeness key performance parameter at 91 percent against a 95 percent threshold. Disqualifying it on paper cost a week of analysis; discovering the same thing after fielding 150 gateways would have cost a recall.

4. Modeling and simulation. Measures of feasibility and performance are estimated before hardware exists, which retires **technical risk at low cost** and converts a design argument into a number. *Example:* a discrete-event simulation of a 150-patient fleet, driven by observed session sizes and measured carrier outage durations, was used to size the gateway buffer. The 72-hour buffer carried in the original concept lost data in roughly 8 percent of simulated months; a 14-day buffer held in all of them. The buffer requirement changed before a single enclosure was tooled.

5. Prototype development of critical components. Explorative prototypes settle requirements with users; experimental prototypes determine whether the planned system will actually be adequate. Both retire **technical risk**, and the experimental prototype can stand in as a requirements specification. *Example:* an experimental HG-1 — a single-board gateway running production firmware and no enclosure — was run against twenty simulated cyclers on a carrier test network for six weeks, with deliberate SIM deactivations and forced firmware faults. It established that store-and-forward survived a nine-day outage and that the planned liveness heartbeat detected a dead link in under 35 minutes.

6. Development test planning and the integration build-up. A test and evaluation plan and a staged integration sequence retire **integration risk**, the risk that the first time elements meet is also the first time anyone learns they do not fit. *Example:* the PD-RPM build-up runs HG-1 alone on the bench; HG-1 with the cycler data port; HG-1 with the Bluetooth peripherals; HG-1 end-to-end to the ingest service over a live carrier; the monitoring-center services against simulated feeds; the full chain with simulated patients; the full chain with clinicians in the loop; and a limited clinical evaluation with real patients last. Each step adds one element, so any defect is attributable to the thing just added.

	Activity	**Risk class retired**	**What it produced for PD-RPM**
1	Identify and rank high-risk elements	Programmatic	HG-1 isolated at TRL 6 as the one element needing the phase
2	Requirements flow-down	Verification	30-minute transfer budget shown to omit carrier re-registration
3	Trade studies	Cost and schedule	Cheapest alternative disqualified on a KPP before procurement
4	Modeling and simulation	Technical	Buffer resized from 72 hours to 14 days before tooling
5	Prototype development	Technical	Store-and-forward verified across a nine-day outage
6	Development test planning	Integration	Eight-step build-up isolating each defect to one element

The common thread is that each activity replaces a belief with a measurement while the cost of being wrong is still small. The phase is called risk reduction precisely because nothing in it is meant to ship: the prototype has no enclosure, the simulation has no patients, the build-up stops short of production. What it delivers is a design the program can commit to and a record of the uncertainties that no longer have to be carried forward.